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DH22TIN07 LESSON 1: Library management -

system - Identify candidate classes (Books, Readers, Loans, Employees, Publishers) + Books: main entity, book information management.

+ DocGia: person who borrows books.

+ PhieuMuon: records borrowing/returning.

+ NhanVien: system manager (librarian, warehouse keeper...).

+ NhaXuatBan: book source information.

- Properties and behavior (methods) of each class + Book

- Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), maNXB (FK), soLuongTon (int), thai page (string: "Available"/"On loan").

- Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho().

+ DocGia

- Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).

- Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon(). + PhieuMuon

- Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), maNhanVien (FK), nhaMuon (date), ngoHenTra (date), ngoTraThucTe (date), tinhTrang (string: "On loan"/"Paid"/"Overdue").

- Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat(). + NhanVien

- Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), date of birth (date), soDienThoai (string).

- Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().()

+ NhaXuatBan

- Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email (string).

- Behavior: addNXB (), capNhatNXB ().

- Relationships: 2 Books and Publishers: 1 book – 1 publisher

- ☐☐ Readers and Borrowing Slips: 1 reader – many borrowing slips
- ☐☐ Books and borrowing slips: 1 book – many borrowing slips
- ☐☐ Employees and Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- ☐☐ DangKyTaiKhan (Customer registration).
- ☐☐ DangNhap (Sign in).
- ☐☐ See SanPham (Search / View product list).
- ☐☐ ThemVaoGioHang (Add to cart).
- ☐☐ DatHang (Order).
- ☐☐ ThanhToan (Online payment / COD).
- ☐☐ View DonHang (View order history).

Admin / NguoiQuanLy:

- ☐☐ QuanLySanPham (Add/Edit/Delete products).
- ☐☐ QuanLyDonHang (View/Confirm/Cancel order).
- ☐☐ QuanLyKhachHang (Customer information management).
- ☐☐ QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- ☐☐ ThanhToan includes XacNhanDonHang.
- ☐☐ DatHang extends ThanhToan (if paying online).
- From use case, find the necessary classes
- ☐☐ Customers (Customer): from register, view orders.
- ☐☐ SanPham (Product): product management.
- ☐☐ GioHang (Cart): temporarily store products before ordering.
- ☐☐ Don Hang (Order): from order, payment.
- ☐☐ ChiTietDonHang (OrderDetail): product details in the order.

☒ ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).

☒ Admin or Personnel (inherited from User if there is a user system).

- Analyze the relationship between 2 classes (specify inheritance if 2 any) □ Customer and Customer: 1 customer – many 2 orders (abstract class) ← inherits 2 ← KhachHang and Admin (KhachHang and 2 Admin are both the same type of User, sharing the same account, password, full name).

Lesson 3: Convert class diagrams into data tables

1. MonHoc table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	